GANDHARVA SCHOOL OF MUSIC • TECHNICAL SPECIFICATION

Relational Database Architecture & Complete Schema Documentation

Comprehensive enterprise data model, entity relationships, security controls, lifecycle state machines, and API integration specifications for the Gandharva live music learning and classroom platform.

SYSTEM TARGET ENGINE
PostgreSQL 15+ / Supabase Cloud

DATA ACCESS LAYER
Prisma ORM Client v7.x (Type-Safe Schema)

DOCUMENT REVISION
Release v2.4 (Production Architecture)

GENERATION DATE
November 2026

SECURITY FRAMEWORK
Role-Based Access Control (RBAC), TOTP 2FA, PII Masking

TIMEZONE CONVENTION
UTC Timestamps (Timestamptz 6) with Dynamic IST Localization

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Confidential • Prepared for Client Architecture Review

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1. System Architecture & Core Data Principles

ARCHITECTURE FOUNDATIONS

The Gandharva School of Music database is architectured to support real-time audio/video WebRTC classrooms, high-concurrency booking, teacher payout ledgers, course enrollments with flexible EMI schedules, student resource sharing, and verified attendance auditing.

CORE ARCHITECTURAL PILLARS

1. Strict Referential Integrity: Enforced via PostgreSQL foreign key constraints with cascading deletes for owned sub-entities (e.g. user tokens, teacher availability) and restrictive/set-null semantics on financial audit rows (e.g. payments, enrollments).
2. Precision UTC Timestamps: All temporal records use `TIMESTAMP WITH TIME ZONE (Timestamptz 6)`. Storage occurs strictly in UTC instants; UI localization renders dynamically into viewer timezones with Indian Standard Time (IST) standardized for admin operations.
3. Dual-Ledger Financial Integrity: Monetary balances, course fees, and teacher payouts are stored as **Minor Units (Integers)** representing Indian Paise (e.g. ₹500 = 50,000 paise). This prevents floating-point rounding errors across payment gateways (Razorpay, UPI) and payout reconciliation.
4. Classroom Auditing & Multi-Join Tracking: Every live lesson maintains both discrete entry/exit logs (`teacherJoinedAt`, `studentJoinedAt`) and comprehensive JSON multi-session timelines in `attendanceLogs`, along with archived in-call chat transcripts in `chatLogs`.

2. Identity, Authentication & RBAC Domain

CORE SECURITY

`model User`

Table: User • 23 Fields • 25 Relations

Core identity record supporting multi-role access control (STUDENT, TEACHER, ADMIN). Handles password hashing, OAuth federated accounts, phone verification, two-factor authentication (TOTP), and demographic profiles.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & BUSINESS RULES
`id`	`String`	PK CUID()	Unique alphanumeric system identifier.
`customId`	`String?`	UNIQUE NULLABLE	Human-readable student ID (e.g. `ST00001`) or teacher ID.
`email`	`String`	UNIQUE REQUIRED	Primary login and communication identifier.
`role`	`Role (Enum)`	STUDENT	System authorization tier: `STUDENT`, `TEACHER`, or `ADMIN`.
`passwordHash`	`String?`	NULLABLE	Bcrypt-hashed credential for credentials-based authentication.
`phone`	`String?`	UNIQUE NULLABLE	E.164 formatted telephone number for SMS notifications and OTP.
`twoFactorEnabled`	`Boolean`	FALSE	Flag indicating if TOTP authenticator app verification is enforced.
`timezone`	`String`	"UTC"	IANA timezone string (e.g. `Asia/Kolkata`, `America/New_York`).
`guardianName / Phone`	`String?`	NULLABLE	Contact details for underage music students.

Primary Relations: 1:1 with `TeacherProfile`, `StudentTrialStatus`; 1:N with `Enrollment`, `Lesson`, `Payment`, `Notification`, `Resource`.

model Account, Session & Tokens

Authentication & Verification Infrastructure

Standards-compliant Auth.js / NextAuth models managing OAuth providers (Google, Apple), persistent JWT/Database sessions, email verification tokens, and OTP phone verification tokens.

MODEL NAME	KEY FIELDS	PURPOSE & SECURITY PROTOCOL
Account	`userId, provider, providerAccountId, access_token`	Federated single sign-on link mapping external OAuth tokens to internal `User`.
Session	`sessionToken, userId, expires`	Active user session tracker with automatic expiration timestamps.
PasswordResetToken	`email, token, expires, used`	Single-use, cryptographically secure 24-hour expiration token for password recovery.
PhoneVerificationToken	`phone, code, expires, used`	Time-bounded 6-digit OTP code for SMS identity verification.

3. Faculty Management & Dynamic Scheduling

TEACHER DOMAIN

model TeacherProfile

Table: TeacherProfile • 21 Fields • 4 Relations

Faculty bio, musical discipline expertise, accreditation, onboarding approval lifecycle, bank/UPI payout credentials, and session remuneration rates.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & BUSINESS RULES
userId	String	FK USER UNIQUE	1-to-1 linkage to the corresponding User record.
expertInstruments	String[]	[]	Primary instruments where teacher holds master/advanced teaching qualifications.
approvalStatus	TeacherApprovalStatus	PENDING	Workflow state: PENDING, APPROVED, or REJECTED.
payoutDayRate	Int	30000 (₹300)	Remuneration in minor units for sessions between 6:00 AM – 8:00 PM IST.
payoutNightRate	Int	50000 (₹500)	Remuneration in minor units for prime/night sessions (after 8:00 PM IST).
upiId	String?	NULLABLE	Direct UPI VPA (e.g. teacher@okhdfcbank) for automated fee disbursement.
isPublished	Boolean	FALSE	Controls visibility on public course booking and instructor catalog pages.

model AvailabilityRule & AvailabilityException

Dynamic Time-Slot Matrix Engine

Powers student self-serve booking and time-zone conversion. Combines weekly recurring availability rules with specific calendar override exceptions.

MODEL	FIELDS	TIME MECHANICS & CONSTRAINTS
AvailabilityRule	teacherId, dayOfWeek, startMinute, endMinute	Weekly repeating slot. dayOfWeek (0 = Sunday to 6 = Saturday). Minutes from midnight (e.g. 540 = 9:00 AM, 1080 = 6:00 PM) stored in teacher's native timezone.
AvailabilityException	teacherId, date, isBlocked, startMinute, endMinute	Calendar-specific date override. isBlocked = true blocks off teacher leave/vacations; isBlocked = false opens special ad-hoc slots.

4. Classroom Sessions, Attendance & Recordings

WEBRTC & CLASSROOM DOMAIN

model Lesson

Table: Lesson • 26 Fields • 5 Relations

Core operational record representing a scheduled or completed 1-on-1 music class. Captures LiveKit room credentials, WebRTC attendance multi-joins, mutual co-presence duration, chat transcripts, and faculty remuneration ledger status.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & BUSINESS RULES
id	String	PK CUID()	Unique lesson ID used as the LiveKit room name.
trackingCode	String?	UNIQUE NULLABLE	Human-readable session tracking ID (e.g. S0017-SE009).
status	LessonStatus	SCHEDULED	Lifecycle: SCHEDULED, RESCHEDULED, COMPLETED, CANCELLED.
lessonSource	LessonSource?	TRIAL | ENROLLMENT	Distinguishes complimentary free trials from paid course sessions.
teacherJoinedAt / LeftAt	DateTime?	TIMESTAMPTZ	Primary classroom entry and exit timestamps for faculty instructor.
studentJoinedAt / LeftAt	DateTime?	TIMESTAMPTZ	Primary classroom entry and exit timestamps for student.
attendanceLogs	Json?	STRUCTURED JSON	Granular timeline array logging every room visit, reconnect, disconnect, and calculated mutual co-presence seconds.
chatLogs	Json?	STRUCTURED JSON	Archived classroom chat transcript including message ID, sender role, sender name, message body, and timestamps.
payoutStatus	TeacherPayoutStatus	UNPAID	Faculty payment status: UNPAID or PAID. Free trials strictly default to UNPAID with 0 amount.
payoutAmountMinor	Int?	NULLABLE	Computed remuneration in paise (e.g. 30,000 paise = ₹300, 50,000 paise = ₹500).

model LessonRecording & LessonRecordingMedia

Cloud Video Archival & Vault Failover

Integrates LiveKit Egress recording with Supabase Storage and Google Drive cloud vault. Provides high-availability video storage for student practice playback and parent quality assurance.

MODEL	KEY FIELDS	STORAGE STRATEGY & ARCHIVAL SEMANTICS
LessonRecording	lessonId, fileUrl, fileSizeBytes, durationSeconds, storageBackend	Metadata record pointing to object storage (Supabase S3 bucket or Google Drive). Stores double-precision file byte size and duration.
LessonRecordingMedia	recordingId, mimeType, bytes	Emergency database vault fallback storing raw binary stream (bytea) in case external cloud storage egress encounters network partitions.

5. Academic Courses, Curricula & Enrollments

ACADEMIC DOMAIN

model Course & CourseLesson

Curriculum Catalog & Syllabus Engine

Defines academic music programs (Carnatic Vocals, Hindustani, Guitar, Piano, Keyboard, Drums, Western Vocals) with accreditation, pricing, EMI plans, and syllabus module roadmaps.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & BUSINESS RULES
title / slug	String	SLUG UNIQUE	Course display name and SEO-friendly unique URL route.
discipline	Discipline	REQUIRED	Classification: INSTRUMENT, VOCALS, or DANCE.
level	CourseLevel	REQUIRED	Academic difficulty: FOUNDATION, BEGINNER, INTERMEDIATE, ADVANCED.
priceMinorUnits	Int	PAISE	Total upfront course fee in minor currency units (e.g. 2400000 = ₹24,000).
emiPrice3/6MinorUnits	Int?	NULLABLE	Monthly recurring installment amounts for 3-month or 6-month EMI plans.
sessionCount	Int	REQUIRED	Total quota of 1-on-1 sessions allotted upon enrollment (e.g. 24 or 48 classes).

model Enrollment

Table: Enrollment • 17 Fields • 6 Relations

The active academic contract binding a student to a course. Tracks remaining session balance, assigned faculty mentor, EMI installment status, and trial conversion pedigree.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & BUSINESS RULES
studentId	String	FK USER	Enrolled student user reference.
courseId	String	FK COURSE	Target curriculum reference.
sessionsRemaining	Int	REQUIRED	Decrementing balance of unbooked class credits available to the student.
status	EnrollmentStatus	ACTIVE	Lifecycle: ACTIVE, COMPLETED, CANCELLED, REFUNDED.
emiStatus	CourseEmiStatus	NOT_APPLICABLE	Payment plan health: FIRST_INSTALLMENT_PAID, PARTIALLY_PAID, FULLY_PAID.
convertedFromTrial	Boolean	FALSE	Analytics flag tracking sales conversion from complimentary trial sessions.

6. Financial Ledger, Invoicing & Certification

FINANCE & CREDENTIALS

model Payment

Table: Payment • 13 Fields • 2 Relations

Immutable double-entry financial transaction log. Integrates with Razorpay Orders and Payment APIs with idempotent webhook confirmation.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & GATEWAY VALIDATION
customId	String?	UNIQUE	Official invoice number (e.g. PAYID000001) issued on fee receipts.
amountMinorUnits	Int	PAISE	Actual captured monetary amount in paise (e.g. 50,000 = ₹500.00).
gatewayOrderId	String	UNIQUE	Razorpay Order ID (order_xxxxxxxxxxxxxx).
gatewayPaymentId	String?	UNIQUE	Razorpay Transaction ID (pay_xxxxxxxxxxxxxx) verified via HMAC SHA256 signature.
status	PaymentStatus	CREATED	Payment state: CREATED, PAID, FAILED, REFUNDED.

model Certificate

Table: Certificate • 6 Fields • 1 Relation

Official graduated credential issued upon successful completion of curriculum modules. Features verifiable serial numbers and vector PDF certificate rendering.

FIELD	TYPE	VERIFICATION & STORAGE LOGIC
certificateNumber	String (Unique)	Publicly verifiable credential code (e.g. GSM-2026-VOC-0042).
pdfUrl	String	Permanent Supabase cloud storage URL pointing to the signed certificate document.
enrollmentId	String (Unique)	Strict 1-to-1 linkage guaranteeing only one certificate per course enrollment.

7. Lead Management, Free Trials & Admissions

CRM DOMAIN

model TrialRequest & StudentTrialStatus

Complimentary Trial Ledger & Admissions CRM

Enforces platform policy on free trials (configured in PlatformSettings, e.g. 2 free trial lessons per student). Manages admissions leads, counselor assignment, and instructor allotment.

FIELD NAME	TYPE	CONSTRAINTS	DESCRIPTION & WORKFLOW RULES
customId	String?	UNIQUE	Lead tracking ID (e.g. TL00001) used in CRM communications.
status	TrialRequestStatus	PENDING	Admissions state: PENDING, ALLOTTED, CANCELLED.
allottedTeacherId	String?	FK USER	Faculty instructor matched and allotted by admissions coordinator.
allottedLessonId	String?	FK LESSON	Direct linkage to the generated 1-on-1 trial classroom session.
isConverted / ConvertedAt	Boolean / DateTime?	FALSE	Flag indicating if the trial student subsequently enrolled in a paid course.

8. In-App Messaging, Safety & Educational Vault

INTERACTION & MATERIALS

model Conversation, DirectMessage & Resource

Child Safety PII Shield & Sheet Music Library

Pairwise student-teacher messaging channel with proactive PII pattern sanitization. Secure sheet music, exercise audio, and homework distribution repository.

MODEL	KEY FIELDS	BUSINESS RULES & SECURITY MEASURES
Conversation	teacherId, studentId, instrument, isActive	Unique 1-to-1 communication channel scoped strictly between an allotted instructor and student.
DirectMessage	conversationId, senderId, messageType, body, fileName, fileUrl	Supports text, file attachments (PDF/DOC only), and homework assignments. Blocked automatically if phone, email, or external social handles are detected.
Resource & Share	teacherId, category, fileUrl, sharedAt, isViewed	Curated sheet music, audio tracks, and lesson notes uploaded to Supabase and shared selectively with enrolled students.

9. Platform Enumeration Dictionary

STATE MACHINES & CONSTANTS

All platform states, user roles, financial flows, and curriculum disciplines are governed by type-safe PostgreSQL enums defined in Prisma.

enum Role

STUDENT TEACHER ADMIN

enum LessonStatus

SCHEDULED RESCHEDULED COMPLETED CANCELLED

enum LessonSource

TRIAL ENROLLMENT

enum Discipline

INSTRUMENT VOCALS DANCE

enum CourseLevel

FOUNDATION BEGINNER INTERMEDIATE ADVANCED

enum EnrollmentStatus

ACTIVE COMPLETED CANCELLED REFUNDED

enum CoursePaymentPlan

FULL_PAYMENT EMI_3_MONTHS EMI_6_MONTHS

enum CourseEmiStatus

NOT_APPLICABLE PENDING_PAYMENT FIRST_INSTALLMENT_PAID PARTIALLY_PAID FULLY_PAID

enum PaymentStatus

CREATED PAID FAILED REFUNDED

enum TeacherApprovalStatus

PENDING APPROVED REJECTED

enum TeacherPayoutStatus

UNPAID PAID

enum TrialRequestStatus

PENDING ALLOTTED CANCELLED

enum ResourceCategory

SHEET_MUSIC AUDIO_LESSON LESSON_NOTES EXERCISE ASSIGNMENT REFERENCE

enum TicketCategory & Status

BILLING TECHNICAL LESSON COURSE OPEN IN_PROGRESS RESOLVED CLOSED

10. End-to-End System Lifecycles & Data Flows

DATA FLOW PIPELINES

LIFECYCLE 1: CLASSROOM SESSION, ATTENDANCE & REMUNERATION PIPELINE

1. **Session Scheduling:** Admin or student schedules class → `Lesson` record inserted with status `SCHEDULED`, unique tracking ID (`S0017-SE009`), and allotted teacher/student.
2. **Live Classroom Entry:** Student & teacher join LiveKit WebRTC room → `/api/lessons/[id]/join` logs entry in `teacherJoinedAt`, `studentJoinedAt`, and appends timestamp session block to `attendanceLogs`.
3. **In-Call Chat & Recording:** Live chat messages persist via `/api/lessons/[id]/chat` into `Lesson.chatLogs`. LiveKit composite egress records video to Supabase S3 storage, creating `LessonRecording`.
4. **Departure & Verified Presence:** Participants leave → `/api/lessons/[id]/leave` records exit. Attendance parser verifies mutual co-presence duration. If session meets qualification, marked `COMPLETED`.
5. **Faculty Payout Ledger:** If paid course session, remuneration computed based on day rate (₹300) vs night rate (₹500). If free trial, remuneration is strictly ₹0 (excluded from payouts).

LIFECYCLE 2: COURSE ENROLLMENT & FINANCIAL LEDGER FLOW

1. **Student Admissions Intake:** Student selects course → creates `CourseEnrollmentRequest` with preferred schedule and payment plan (Full, 3-Month EMI, or 6-Month EMI).
2. **Razorpay Order Dispatch:** Order created via Razorpay API → `Payment` record created with status `CREATED`, unique order ID, and amount in minor units (paise).
3. **Payment Webhook Verification:** Razorpay webhook verifies HMAC signature → updates `Payment.status = PAID`, sets `gatewayPaymentId`, and generates custom invoice ID.
4. **Academic Contract Provisioning:** `Enrollment` record generated with `sessionsRemaining = Course.sessionCount`, status `ACTIVE`, and linked teacher allotted.

End of Specification • Gandharva School of Music Database Architecture Document • Confidential